Ab Initio Interpretation of the Isospin-Symmetric Island of Inversion

Xingcan Cao and Changfeng Jiao

School of Physics and Astronomy, Sun Yat-sen University

Outline

1. Introduction

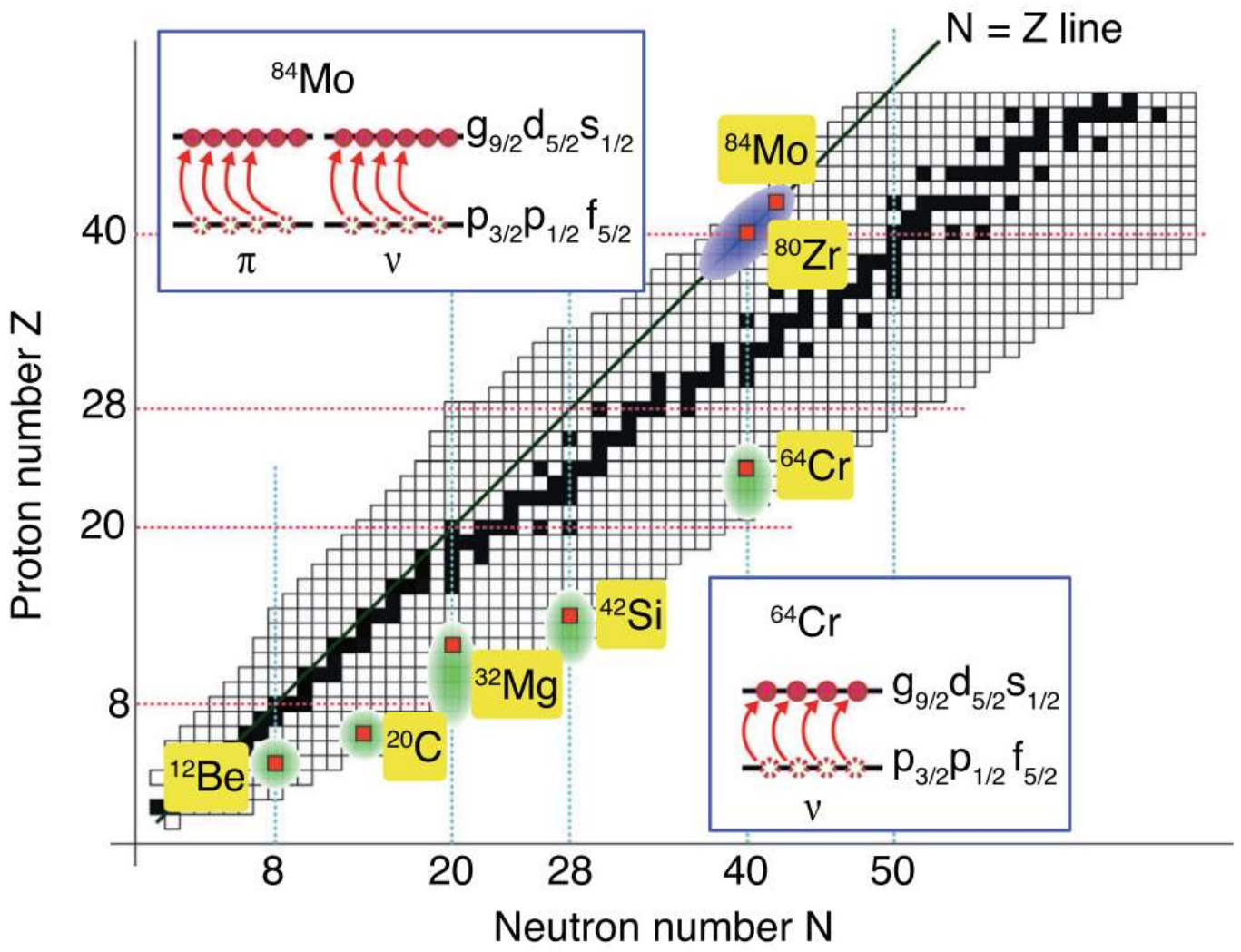
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4. Summary

Introduction

- Rapid structural evolution near $N, Z \sim 40$
- Recent E2 measurements in the N≈Z≈40 region reveal an "isospin-symmetric island of inversion,"
- Experimental signatures are highly non-monotonic
- The microscopic origin remains under debate



J. Ha *et. al.*, Nature Communications 16, 10631 (2025)

Introduction

Table 1 | Computed B(E2; $2_1^+ \rightarrow 0_1^+$) values

		B(E2; $2_1^+ \rightarrow 0_1^+$) (e^2fm^4)				
nuclide	*np-nh*	QPSU3	PHF	DNO-SM	SM	Exp
^{84}Mo	4p-4h	1104	1121	1512	-	1740^{+580}_{-430}
	8p-8h	1891	1794			
^{86}Mo	0p-0h	424	196	893	731	707(71)
	2p-2h	1143	871			
	4p-4h	1416	1179			
	6p-6h	1858	1655			

The values are obtained using different models: quasi/pseudo-SU(3) estimates (QPSU3) and Projected Hartree-Fock solutions (PHF) for different *np-nh* configurations, DNO shell-model calculations (DNO-SM), shell-model diagonalization (SM), and experimental values (Exp) for ^{84,86}Mo.

J. Ha *et. al.*, Nature Communications 16, 10631 (2025)

Using realistic effective interaction DNP-ZBM3.

$B(E2; 2_1^+ \rightarrow 0_1^+)$ (e^2 fm^4) for ^{88}Ru

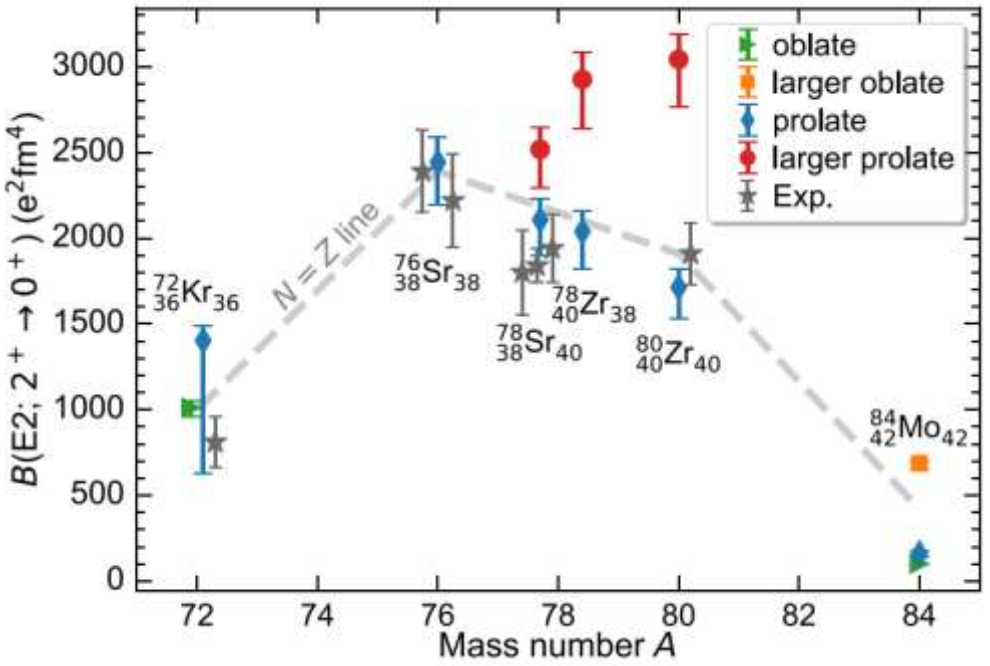
$np-nh$	QPSU3	PHF	LSSM	DNO-SM(VAP)	Exp
0p–0h	833	563	558	616	640^{+200}_{-100}
2p–2h	1185	781			
4p–4h	1600	1299			
6p–6h	2048	2147			

M.A. Bentley *et. al.*, Phys. Rev. Lett. 136,152501 (2026)

- **Phenomenological defects：**
- **Empirical fitting to data on finite nuclei.**

Introduction

Angular-momentum-projected coupled-cluster calculations marked an important *ab initio* advance, but did not simultaneously reproduce rotational spectra and quadrupole strengths for all nuclei near the maximum of collectivity.



B. S. Hu *et. al.*, PRC 110, L011302 (2024)

We employ the ab initio VS-IMSRG+PGCM framework to uncover the microscopic interplay between shell evolution, cross-shell excitations, and the emergence of collectivity in the N=Z region from ^{72}Kr to ^{88}Ru.

Outline

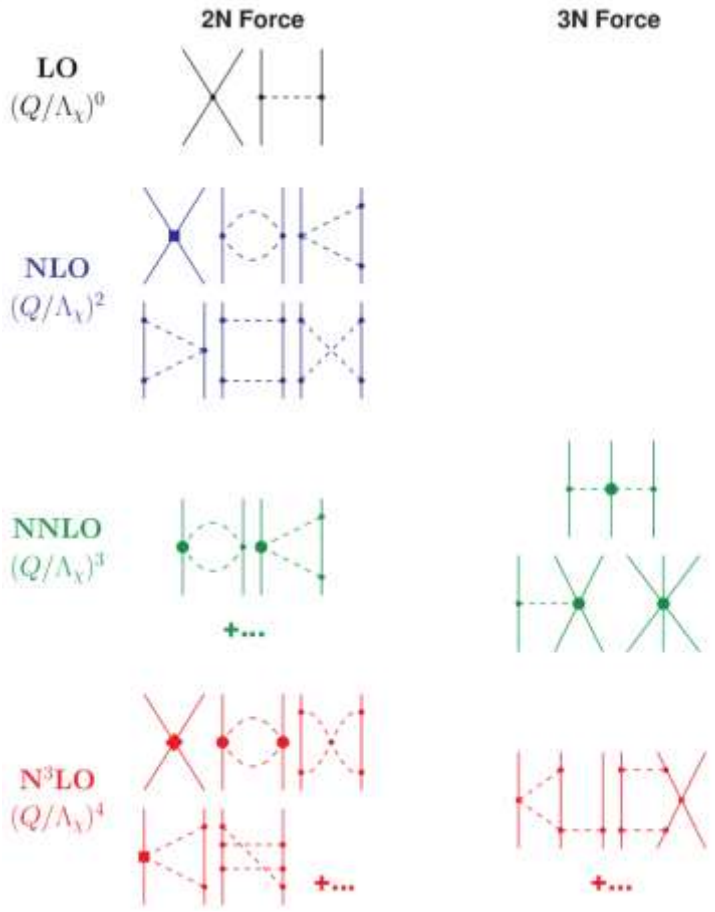
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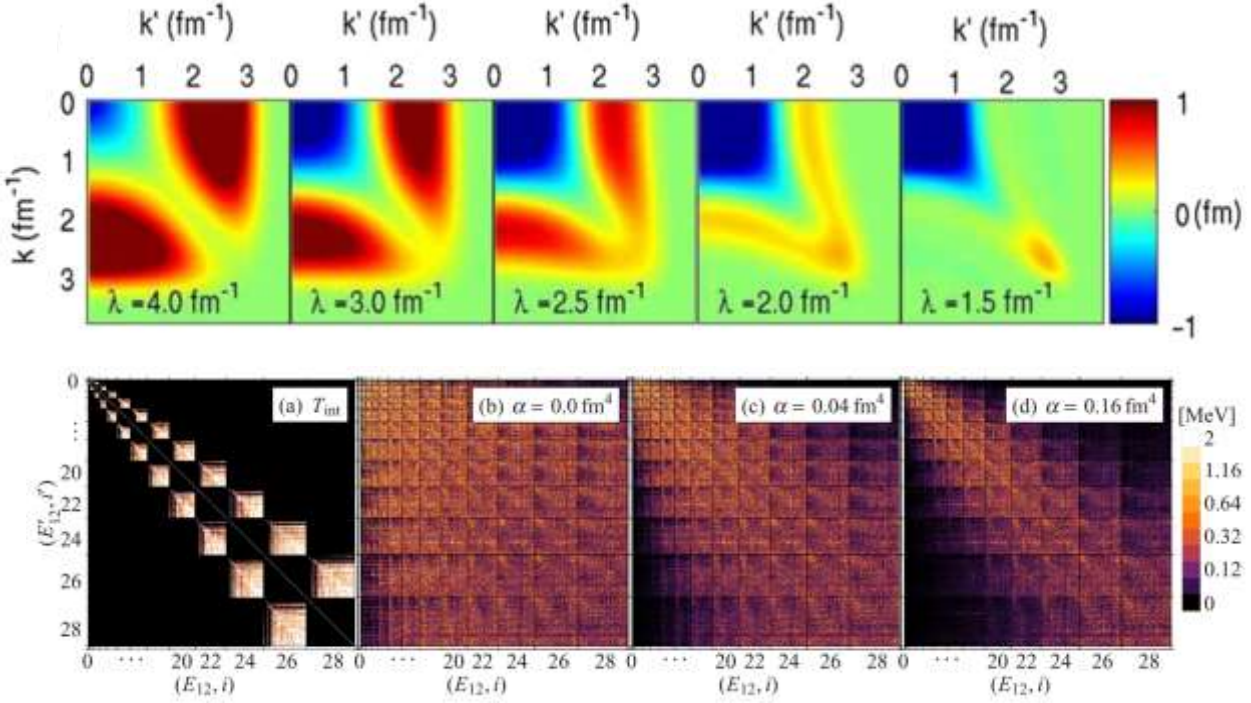
4. Summary

Chiral effective field theory nuclear force



R. Machleidt *et al.*, Phys. Rep. 503, 1-75 (2011)

Soften nuclear force.



R. Roth *et al.*, PRC 90, 024325 (2014)

Expanding in harmonic oscillation (HO) basis.

Methodology: ab initio multi-shell VS-IMSRG Hamiltonian

So we turn to Valence-space In-Medium Similarity Renormalization Group (VS-IMSRG)

We start from the chiral NN+3N forces. The effective Hamiltonian can be expressed in terms of a unitary transformation ***U*** acting on the initial Hamiltonian

$$H_{\text{eff}} = UHU^{\dagger}.$$

It is parameterized by a continuous flow parameter *s* and applied to the Hamiltonian through the flow equation

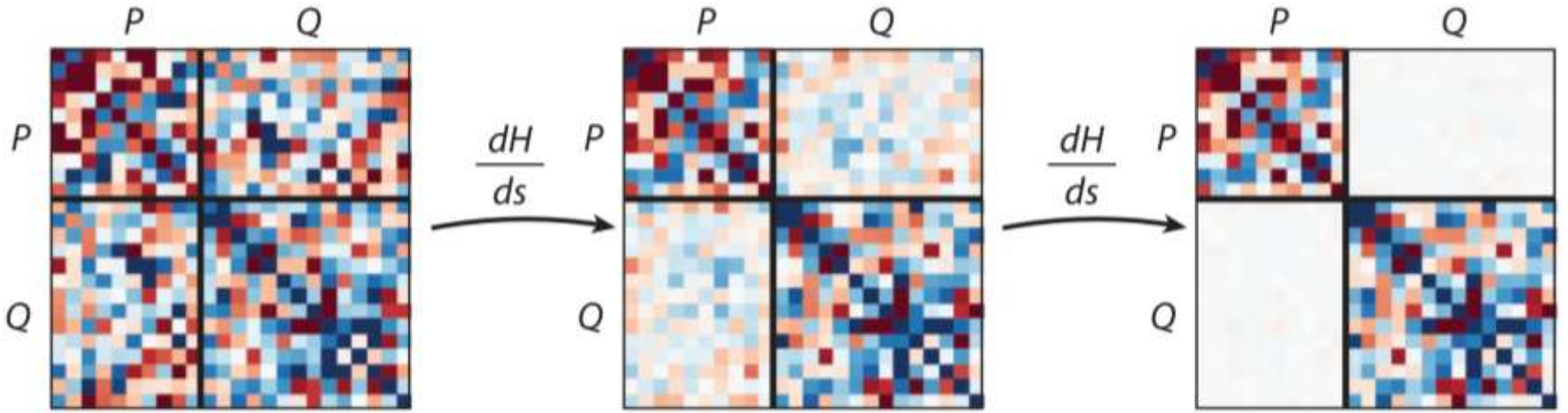
$$\frac{dH(s)}{ds} = [\eta(s), H(s)], \qquad \eta(s) \equiv \frac{dU(s)}{ds}U^{\dagger}(s) = -\eta^{\dagger}(s).$$

Our goal:

$$\lim_{s\to\infty} H^{od}(s) = 0$$

and therefore

$$\lim_{s\to\infty} H^{d}(s) = H_{\text{eff}}.$$



Methodology: ab initio multi-shell VS-IMSRG Hamiltonian

Valence-space In-Medium Similarity Renormalization Group (VS-IMSRG)

The anti-Hermitian generator for VS-IMSRG is

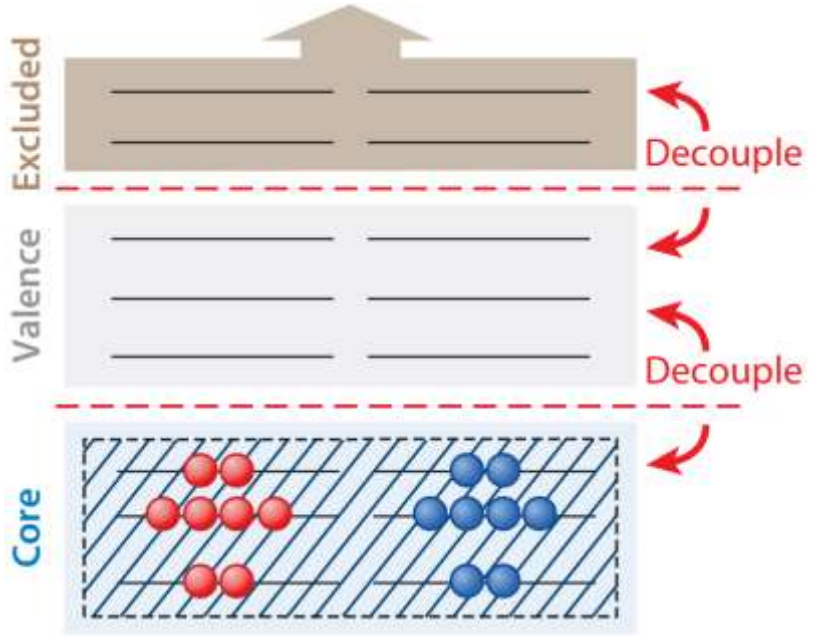
$$\eta = \sum_{ai} \eta_{ai} \{a_a^\dagger a_i\} + \sum_{abij} \eta_{abij} \{a_a^\dagger a_b^\dagger a_j a_i\} - \text{H.c.} \qquad ai \in \{pc, ov\}, \quad abij \in \{pp'cc', pp'vc, opvv'\}.$$

The indices *c*, *v*, and *o* indicate core, valence, and outside space orbitals, respectively, and *p* denotes either *v* or *o*.

$$\eta_{ai} = \frac{1}{2} \arctan\left(\frac{2 f_{ai}}{f_{aa} - f_{ii} + \Gamma_{aiai} + \Delta} \right),$$

$$\eta_{abij} = \frac{1}{2} \arctan\left(\frac{2 \Gamma_{abij}}{f_{aa} + f_{bb} - f_{ii} - f_{jj} + G_{abij} + \Delta} \right),$$

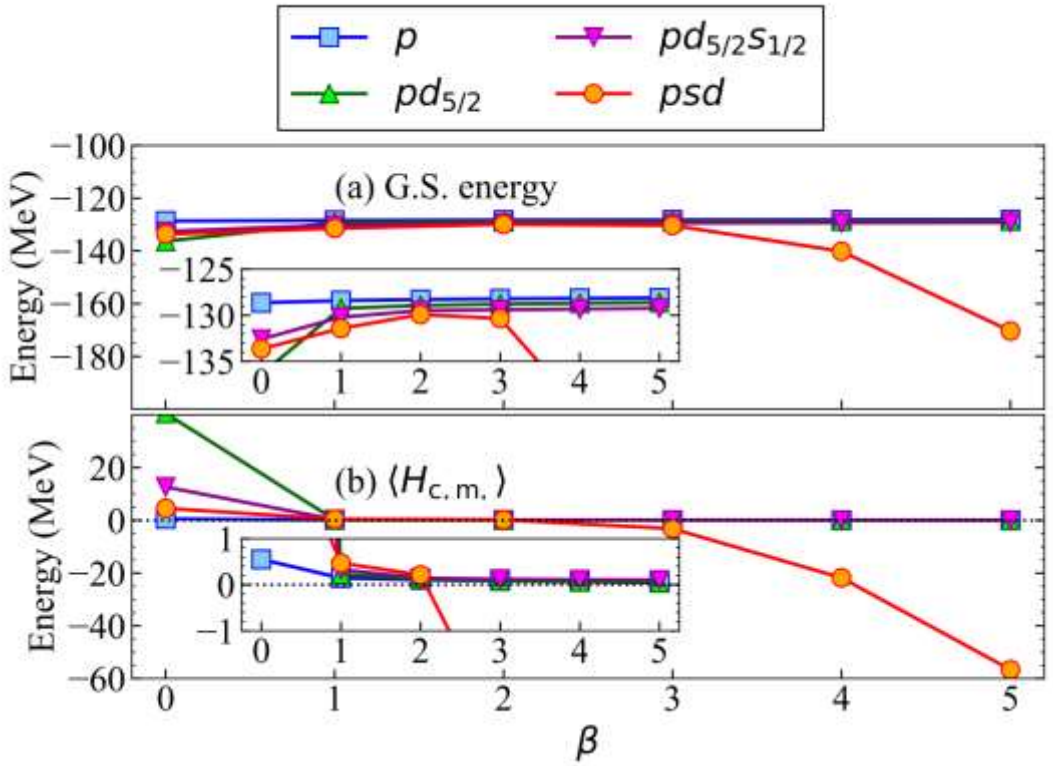
$$G_{abij} = \Gamma_{abab} + \Gamma_{ijij} - \left(\Gamma_{aiai} + \Gamma_{bjbj} + [a \leftrightarrow b] \right).$$



Methodology: ab initio multi-shell VS-IMSRG Hamiltonian

In multishell valence-space Hamiltonians, the issue of center-of-mass (c.m.) contamination must be treated carefully. Our initial Hamiltonians is

$$H = H_{\text{intr}} + \beta H_{\text{c.m.}} \qquad H_{\text{c.m.}} = \mathbf{P}^2/2Am + mA\tilde{\omega}^2\mathbf{R}^2/2 - 3\hbar\tilde{\omega}/2$$



the expectation value $\langle H_{c.m.} \rangle$ should vanish

TABLE I. The scaling factor β and the c.m. Hamiltonian expectation values $\langle \hat{H}_{\text{c.m.}} \rangle$ (in MeV) for 0_1^+ and 2_1^+ states given by PGCM configuration mixing.

Nuclei	β	$\langle \hat{H}_{\text{c.m.}} \rangle_{0_1^+}$	$\langle \hat{H}_{\text{c.m.}} \rangle_{2_1^+}$
^{72}Kr	1	−0.14	−0.43
^{76}Sr	4	−0.73	−0.72
^{80}Zr	3	−1.13	−1.19
^{84}Mo	4	0.02	−1.39
^{86}Mo	4	−0.25	−0.32
^{88}Ru	3	−0.32	−0.61

T. Miyagi *et. al.*, Phys. Rev. C 102. 034320 (2020)

Methodology: ab initio multi-shell VS-IMSRG Hamiltonian

Magnus formulation for VS-IMSRG

Express the IMSRG transformation as the true exponential of the Magnus operator

$$H(s) = e^{\Omega(s)} H(0) e^{-\Omega(s)} = H(0) + [\Omega(s), H(0)] + \frac{1}{2}[\Omega(s), [\Omega(s), H(0)]] + \ldots .$$

we use the BCH formula to obtain

$$\Omega(s+ds) = \Omega(s) + \eta(s)ds + \frac{1}{2}[\eta(s), \Omega(s)]ds + \frac{1}{12}[\Omega(s), [\Omega(s), \eta(s)]]ds + \ldots . \quad \textbf{or} \quad \frac{d\Omega}{ds} = \sum_{k=0}^{\infty} \frac{B_k}{k!} \mathrm{ad}_{\Omega}^{(k)}(\eta),$$

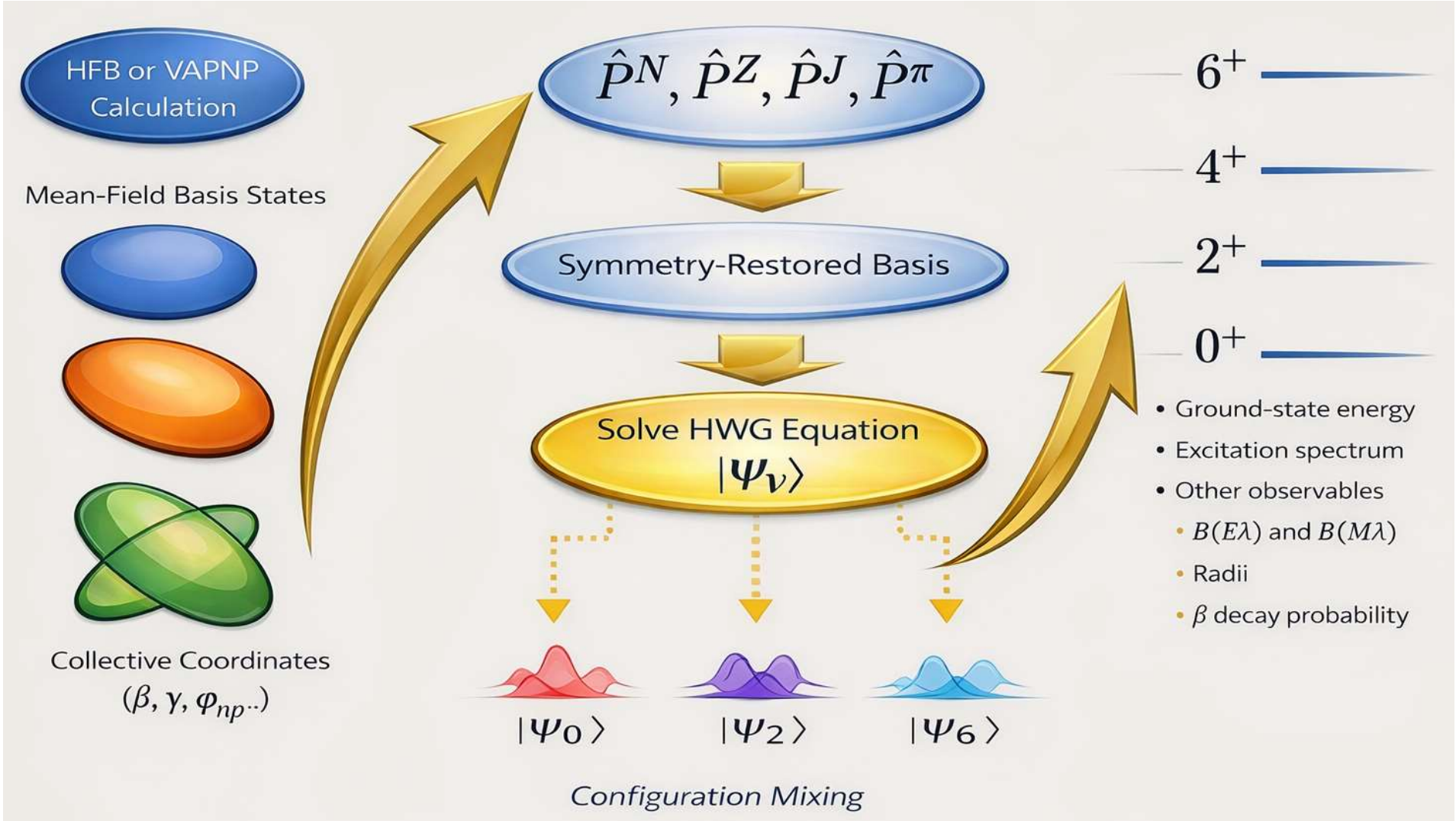
By solving for $\Omega(s)$, we can compute arbitrary effective operators in a consistent VS-IMSRG transformation.

$$\mathcal{O}_{\mathrm{eff}} = e^{\Omega} \mathcal{O} e^{-\Omega} = \mathcal{O} + [\Omega, \mathcal{O}] + \frac{1}{2}[\Omega, [\Omega, \mathcal{O}]] + \ldots .$$

Cheers! We don't rely on effective charges anymore!

A possible way to extend to larger multi-shell spaces: PGCM

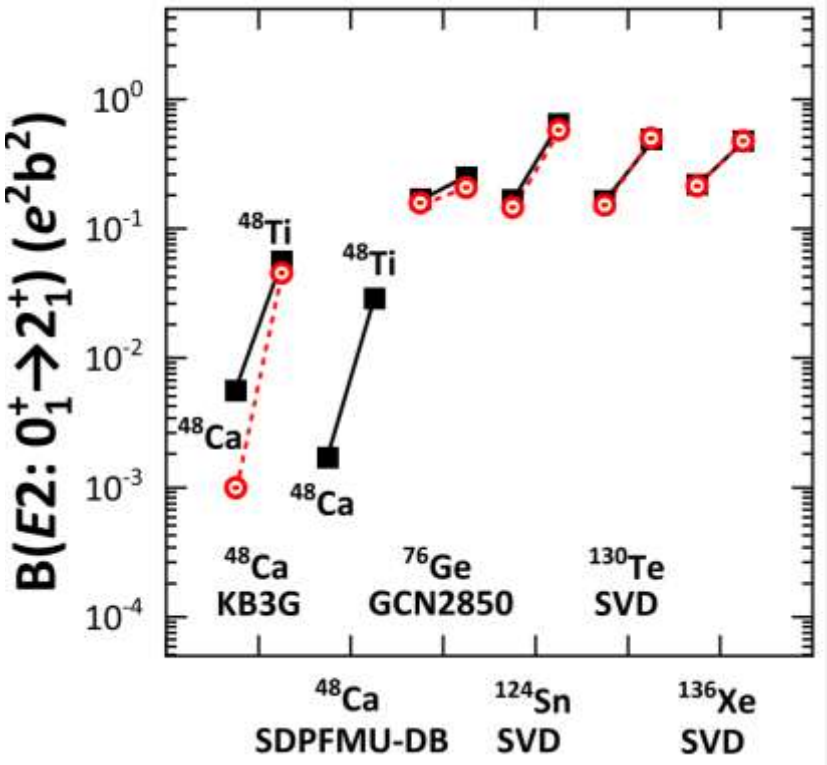
PGCM: A good variational approximation of exact diagonalization in multishells.



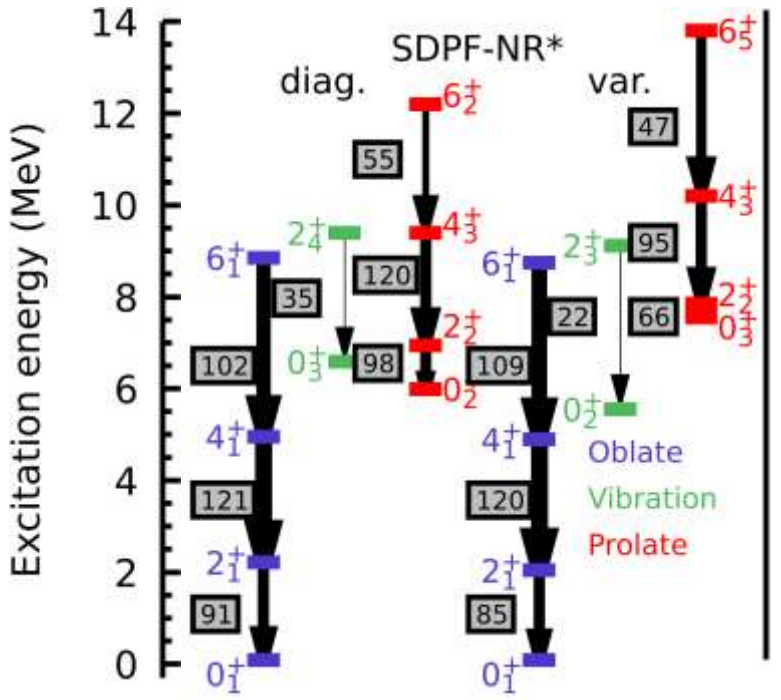
- **Makes shell-model-like calculations in large valence spaces feasible.**
- **VS-IMSRG +PGCM is suitable for *ab initio* studies on heavy open-shell nuclei.**

A possible way to extend to larger multishell spaces: PGCM

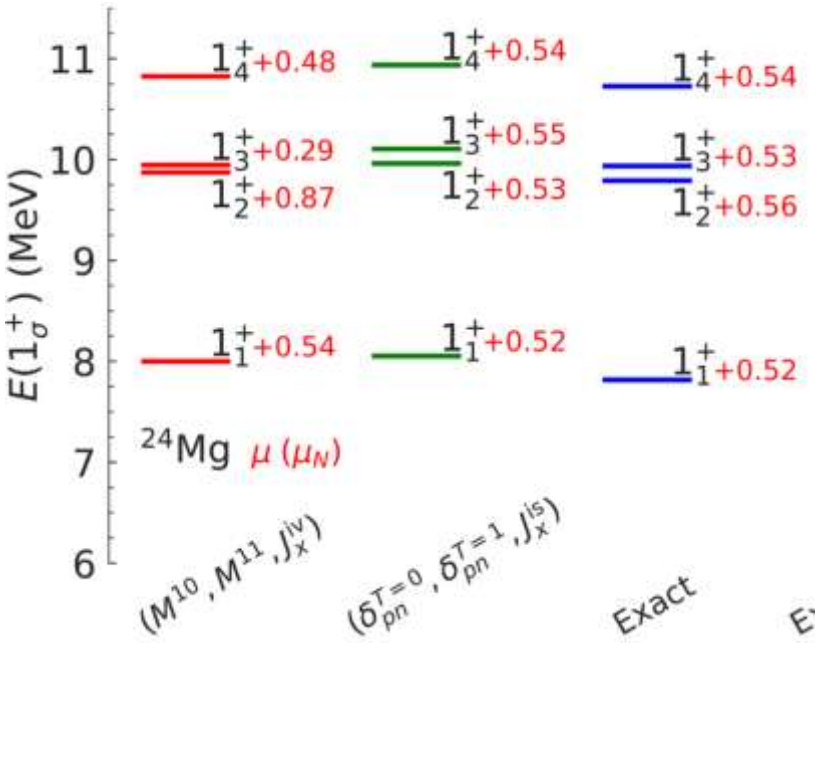
PGCM: A good variational approximation of exact diagonlaization in multishells.



CFJ *et al.*, Symmetry 15, 552 (2023)



Frycz *et. al.*, PRC 110, 054326 (2024)



Bofos *et. al.*, PRC 112, 064312 (2025)

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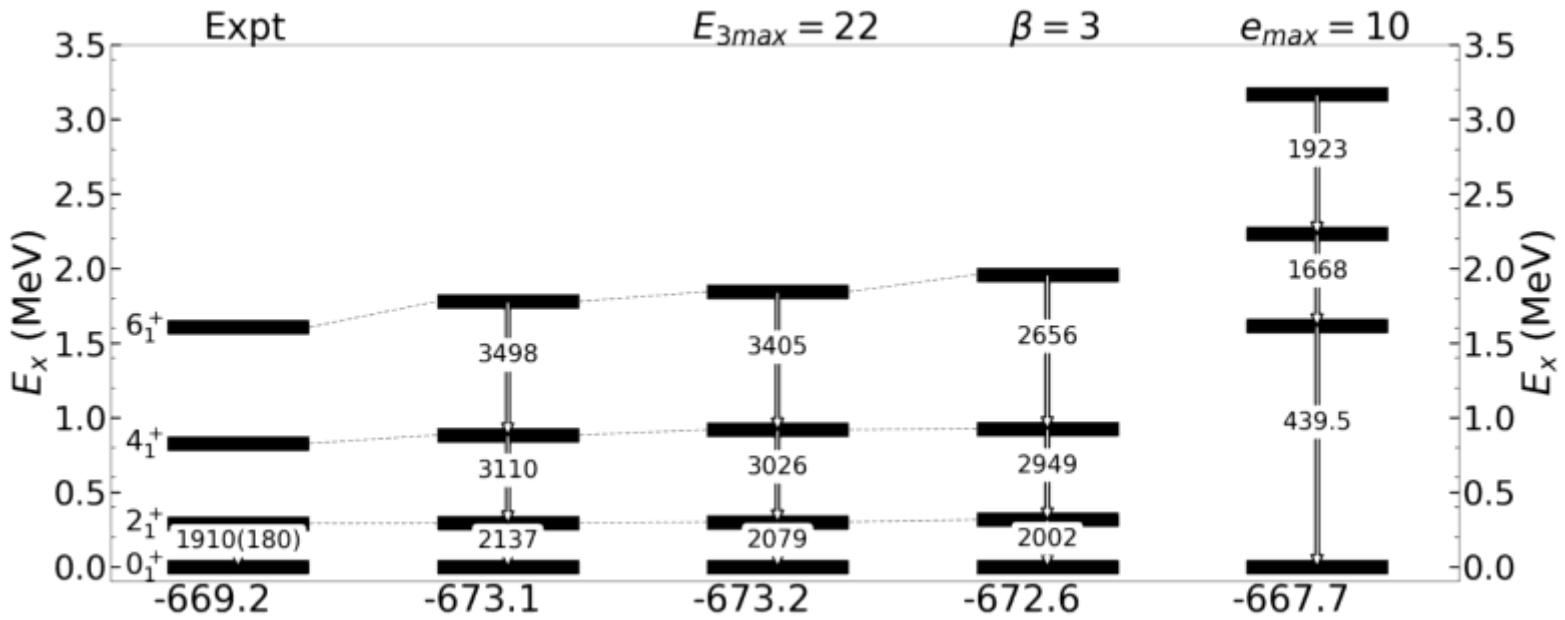
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Results

^{80}Zr:



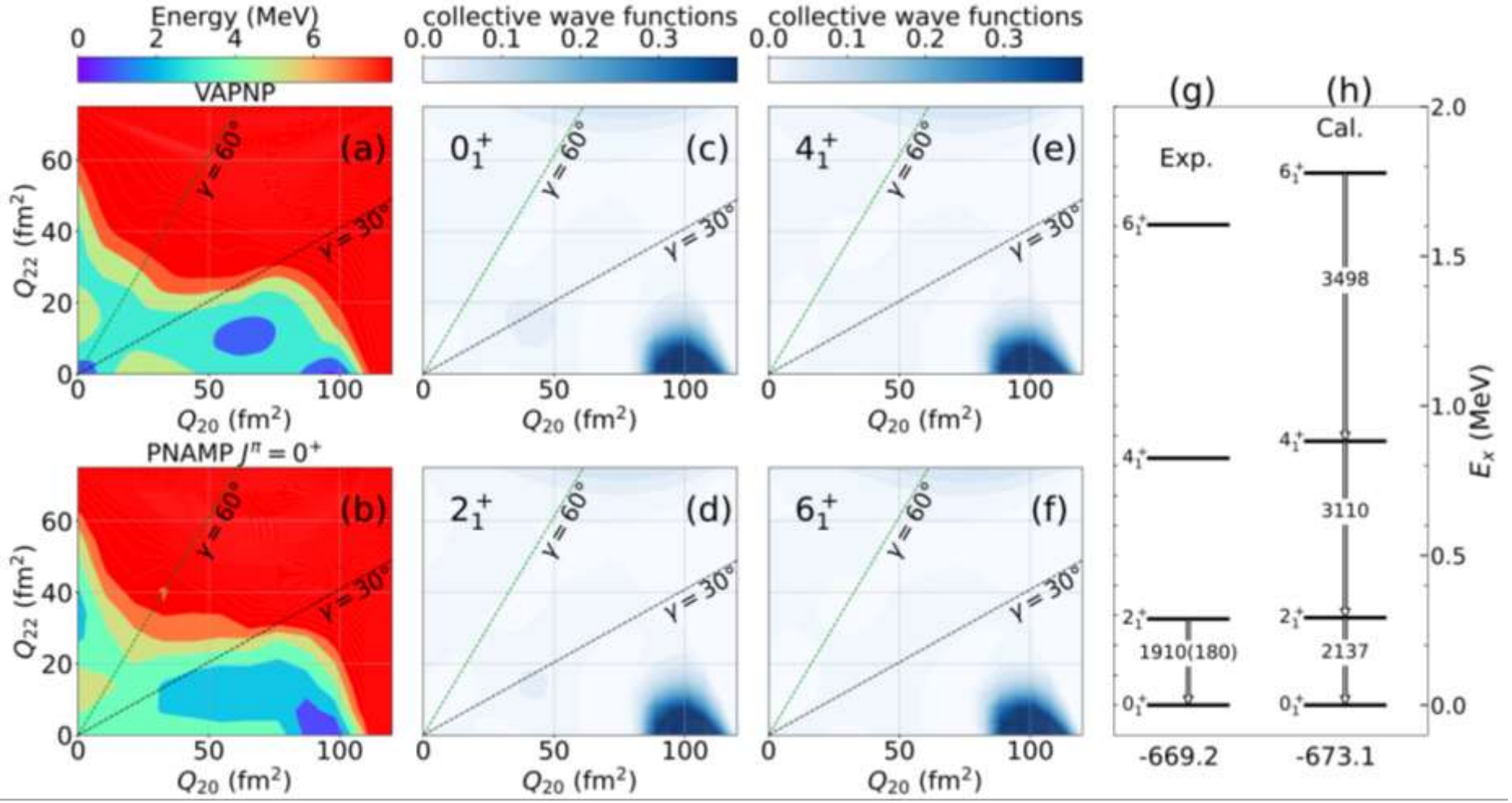
$\hbar\omega$=16, e_{max}=12,E_{3max}=24, β=4

valence space (^{56}Ni core) :
proton and neutron orbitals (28-82):
{0f5/2; 1p3/2; 1p1/2; 0g9/2; 1d5/2;
0g7/2; 2s1/2; 1d3/2; 0h11/2}

- The dependence of on constrain of E_{3max} and β is vary small.

Results

^{80}Zr:



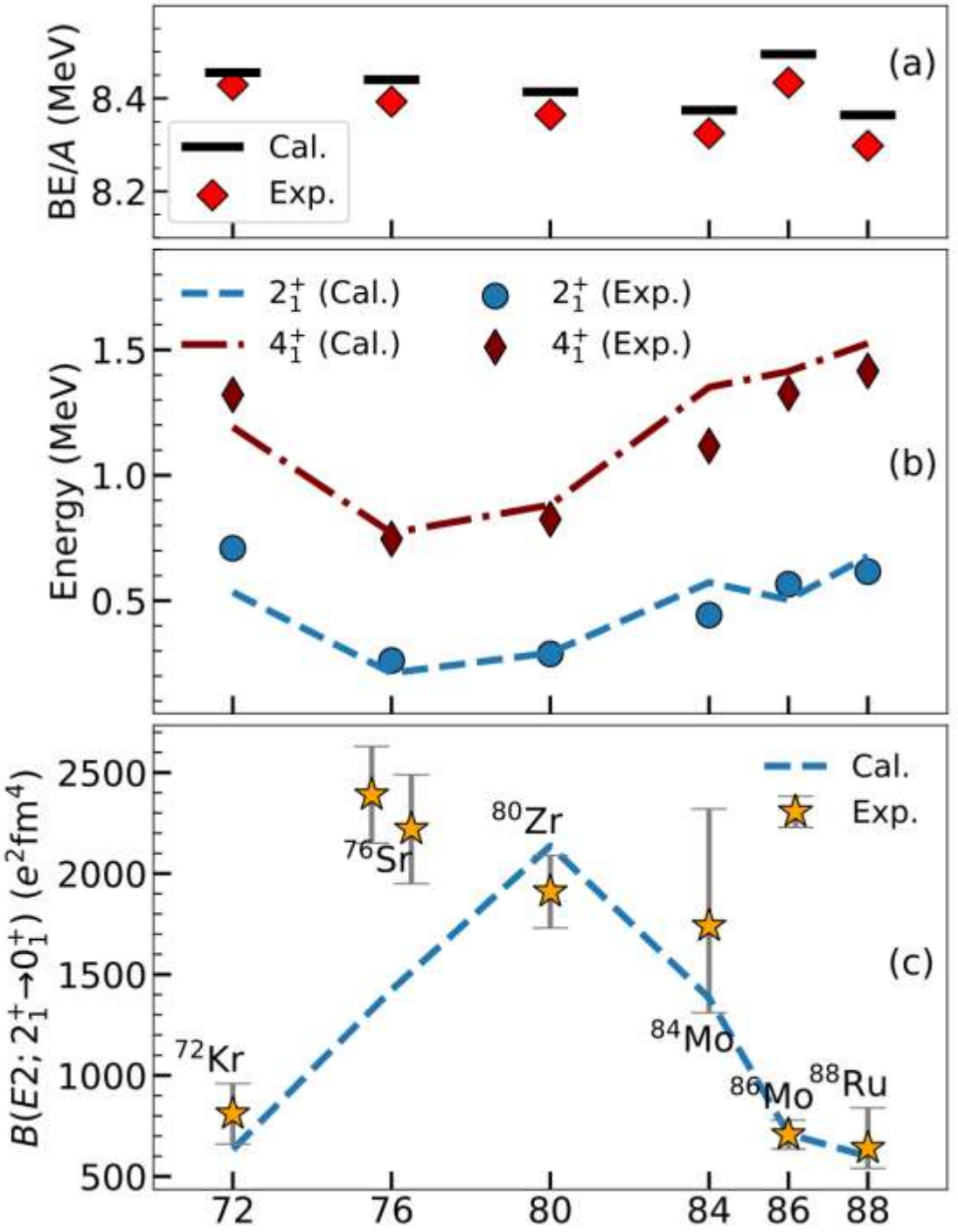
- VAPNP energy surface exhibits three nearly degenerate coexisting minima, after particle-number and angular-momentum projection, prolate minimum gains the largest symmetry-restoration energy.
- We reproduces the observed low-lying rotational band and ground-state energy.

Results

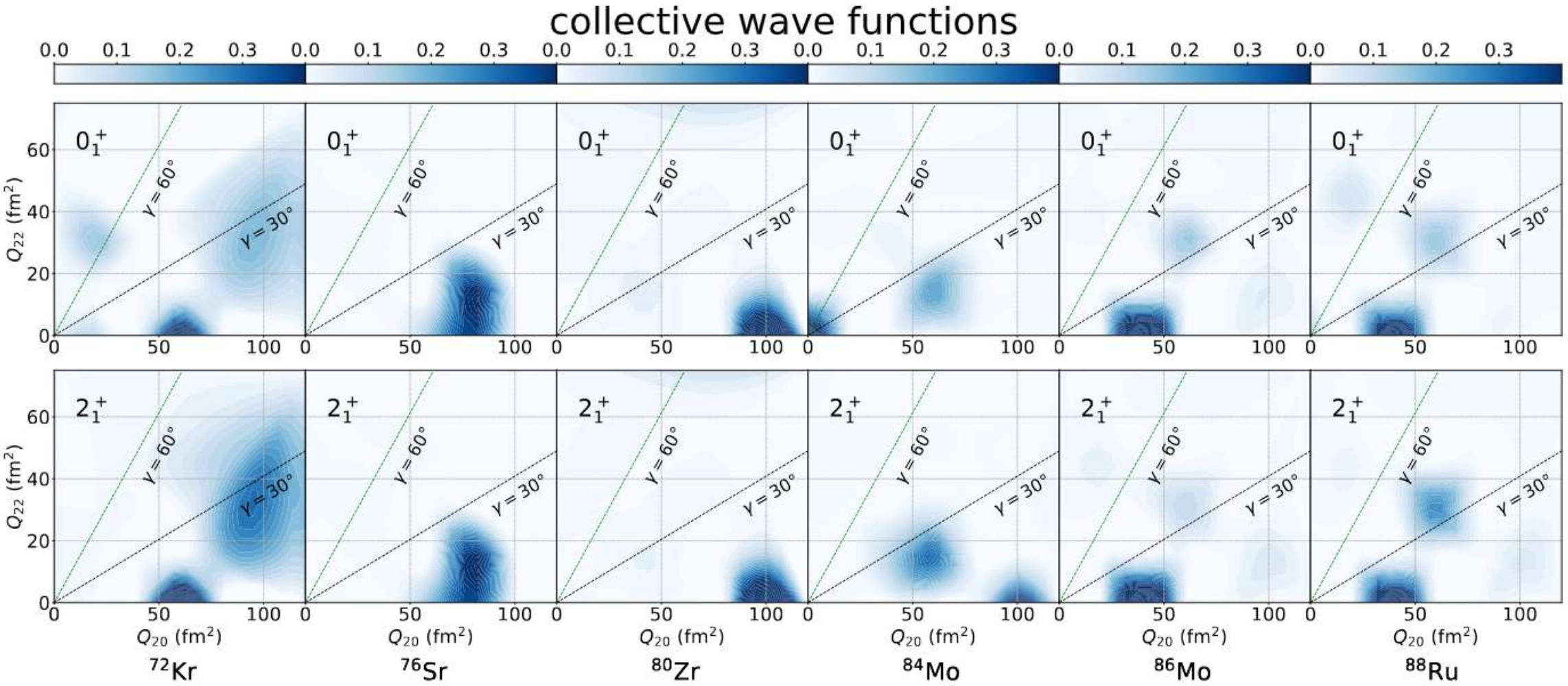
We extend the same calculation from ^{80}Zr to the full $N = Z$ island.

- We reproduce the binding energies, low excitation energy and $E2$.
- The calculations capture the observed nonmonotonic trend in spectroscopic observables, with low excitation energies and large $E2$ strengths in ^{76}Sr, ^{80}Zr, and ^{84}Mo.

To understand the microscopic origin of this behavior, we now examine the collective wave functions of the low-lying states.



Results



A significant shape evolution is observed, from a weakly prolate configuration in ^{72}Kr to strongly prolate shape in ^{76}Sr and ^{80}Zr, and then decreases again toward smaller deformations in the heavier nuclei.

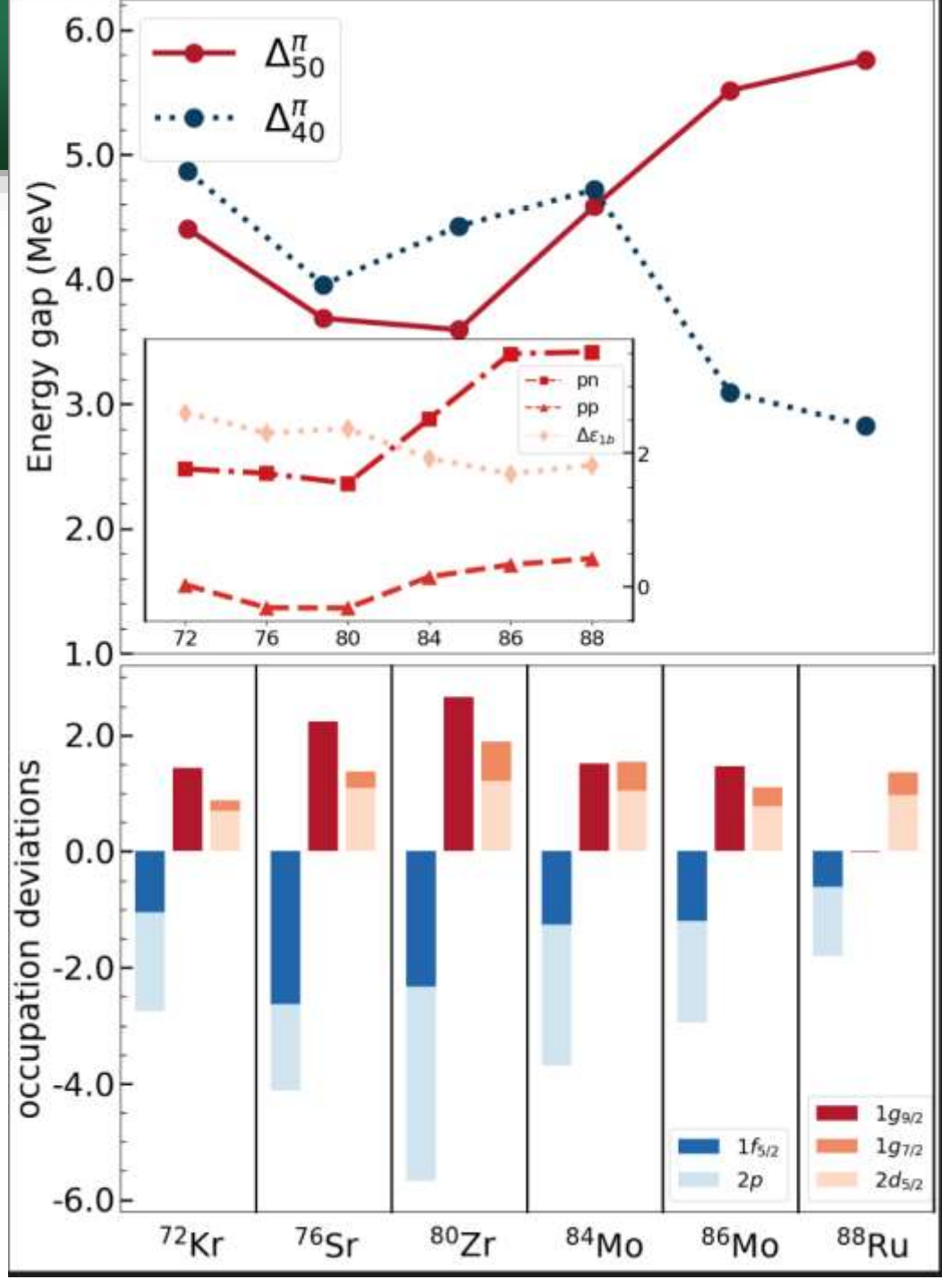
Results

To uncover the microscopic mechanism behind the non-monotonic collectivity, we now investigate the shell evolution through the effective single-particle energies (ESPEs) and the associated particle-hole excitations.

$$\epsilon_i^\pi = \varepsilon_{1b,i}^\pi + \sum_j \langle n_j^\pi \rangle \, \bar{V}_{ij}^{\pi\pi} + \sum_j \langle n_j^\nu \rangle \, \bar{V}_{ij}^{\pi\nu}$$

- The $Z = 40$ ESPE gap remains open in ^{76}Sr and ^{80}Zr, the $1g_{9/2}$– $2d_{5/2}$ spacing decreases to about 3 MeV in these nuclei.

- The reduced $1g_{9/2}$–$2d_{5/2}$ spacing in ^{76}Sr and ^{80}Zr favors simultaneous occupation of the $1g_{9/2}$ and $2d_{5/2}$ orbits, while holes develop in the pseudo-SU(3) *pf* shell.

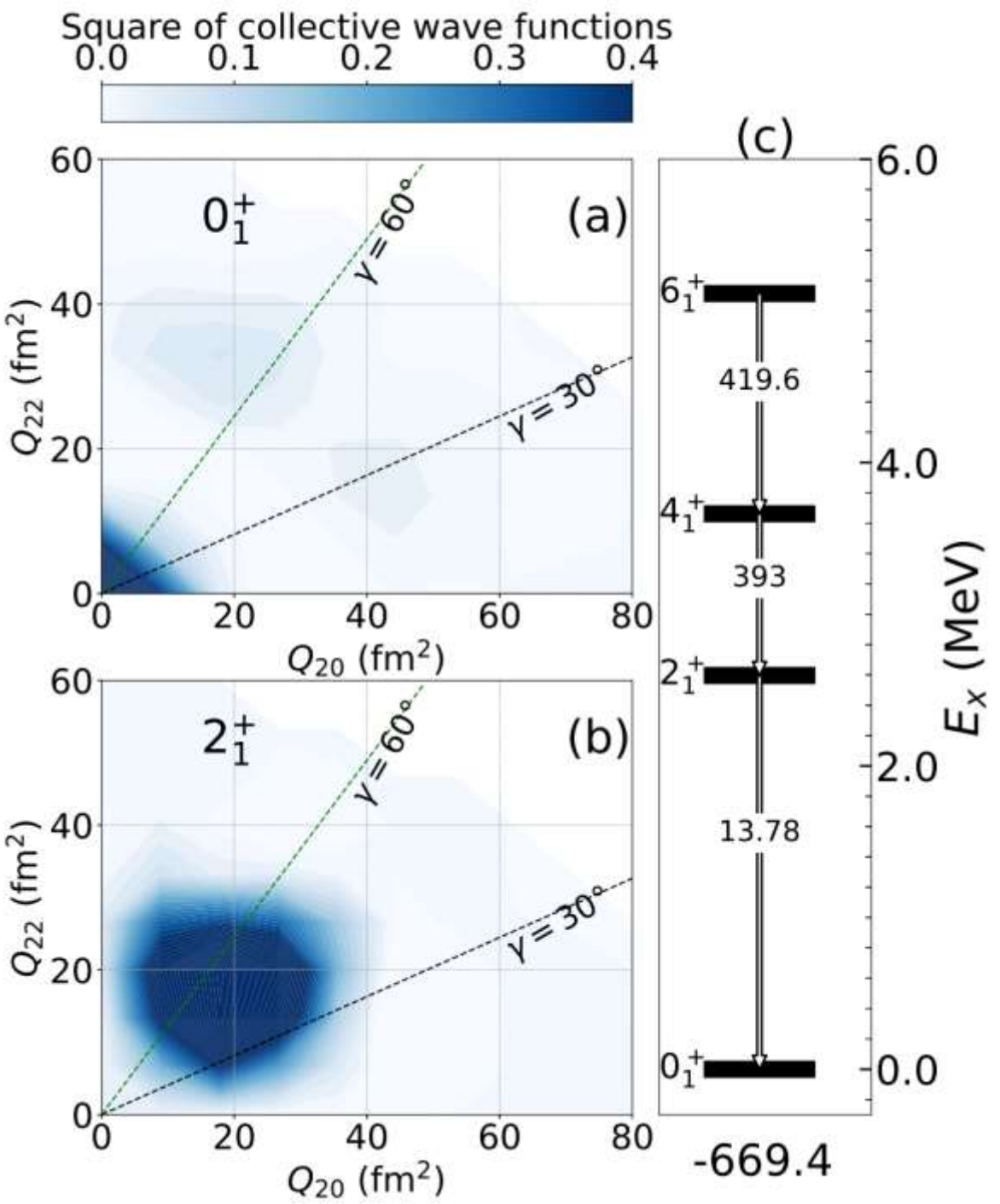


Results

To verify the proposed microscopic mechanism, we perform a constrained calculation by forcing the **occupation of the proton and neutron $1d_{5/2}$ orbital to zero.**

- The ground-state collective wave function shifts from the prolate region to near-spherical deformation, the 2_1^+ state becomes weakly triaxial.
- The yrast spectrum is stretched, and the *E*2 strength is strongly reduced.

The collectivity therefore cannot be generated by *pf*-shell holes and $1g_{9/2}$ occupation alone.



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Conclusions

- Successfully reproduced the non-monotonic evolution of the low-lying spectra and B(E2) strengths.
- Identified the microscopic mechanism through the evolution of ESPEs and cross-shell particle-hole excitations.
- Cross-shell excitations involving the proton 1d5/2 orbital provide the microscopic mechanism for the emergence of strong deformation around the nominal N=Z=40 shell closure.

Thanks for your attention!